

Guidelines for Improving Pain Management and anxiety associated with Intravenous Cannulation and Venepuncture in the Children's Emergency Department

Managing pain and anxiety effectively during intravenous (IV) cannulation and venepuncture is especially important in paediatric patients, as these procedures can cause significant distress, anxiety, and discomfort in both the child and the accompanying caregiver. Children may have heightened sensitivity to pain and may struggle to understand or cooperate with the procedure. In addition, difficult experiences they may have encountered in the Children's Emergency Department can negatively impact future blood taking/cannulation encounters e.g. in the outpatient department/ward.

To ensure that paediatric patients experience as little discomfort as possible, here are evidence-based guidelines focused on pain management for such needle procedures.

These guidelines can be applied to children up to the age of 10 years. Most children above the age of 10 years tend to tolerate needle procedures better, however should this not be the case, we can apply these guidelines to the older cohort as well.

(Appendix B has an easy to follow flow chart for quick reference)

1. Pre-Procedure Preparation and Communication

Developmentally Appropriate Communication:

- Tailor the explanation of the procedure according to the child's age and understanding. For younger children, use simple, reassuring language, and for older children, provide more detailed explanations.
- Be honest about the procedure, but avoid exaggerating the pain. Offering reassurance about the temporary nature of discomfort can reduce fear.

2. Topical Anaesthesia and Pain Relief

• Topical Anaesthetic Creams:

There are a number of different anaesthetic creams used globally to help numb the area for iv cannulation/venepuncture prior to procedure. At NUH, we have EMLA (Eutectic Mixture of Local Anaesthetics) available.

- EMLA cream is a useful and safe topical anaesthetic containing a mixture of lidocaine (2.5%) and prilocaine (2.5%) that can be applied to the skin prior to needle procedures such as iv cannulation and venepuncture.
- EMLA can be used in children of all ages but for children younger than 1 year, it should be used with caution due to the potential for systemic absorption.
- For children over 1 year, EMLA is generally safe when applied according to the manufacturer's guidelines.

EMLA should not be used in:

- Children with known hypersensitivity or allergy to lidocaine, prilocaine, or other local anaesthetics.
- Areas of skin that are broken or infected (e.g., wounds, rashes).
- Children with methemoglobinemia as prilocaine can exacerbate it.
- Infants younger than 12 months should avoid use on large areas of the body or for prolonged periods due to the increased risk of systemic toxicity.

Application of EMLA Cream:

- EMLA should be applied 60 minutes prior to the procedure. (It can remain in place for up to 2 hours for optimal effect.)
- Note down the time of application.
- Early identification of children who may need iv cannulation/venepuncture will help to speed up the process in CE, therefore avoiding lengthening stay in CE where possible.
- Apply a thick layer of EMLA cream (approximately 1-2 mm thick) directly to the site where the cannulation or venepuncture will occur. Common sites include the dorsum of the hand, forearm, or antecubital fossa.
- Cover the cream with an occlusive dressing, such as Tegaderm, to enhance absorption and prevent contamination.
- Ensure that the dressing is secure to prevent the child from removing it before the procedure
- Prior to cannulation/venepuncture, remove the dressing and wipe off any excess cream.

Side effects of EMLA:

- Advise parents or caregivers on the importance of monitoring for any signs of adverse reactions at the application site, such as redness, swelling or blistering. Mild redness or irritation may occur at the application site, but these reactions typically resolve once the cream is removed.
- If any signs of severe allergic reactions (such as swelling, rash, or difficulty breathing) are observed, the procedure should be postponed and the allergic reaction managed accordingly.
- Methemoglobinemia: Although rare, methemoglobinemia can occur, particularly with high doses of prilocaine. Symptoms to watch for include cyanosis (bluish discoloration of the skin), difficulty breathing, and lethargy. In the event of suspected methemoglobinemia, immediate medical intervention is necessary.
- **Vapocoolant Sprays (Cold spray):**
 - **Vapocoolant sprays** (such as ethyl chloride) can rapidly cool the skin and reduce the sensation of pain when the needle is inserted, offering immediate but temporary relief.
 - This method is particularly useful for children who have an aversion to topical creams or require more immediate pain relief.

3. Comfort and Non-Pharmacological Strategies

- **Parent or Caregiver Support:**

- Having a parent hold the child in a secure and comforting manner can help reduce distress. For very young children, the **parent's physical presence** can provide a sense of safety and calm.
- Comfort positioning on the parent's lap, e.g. chest to chest versus lying the child supine is preferable and less intimidating for the child. (see Appendix A for examples of comfort positioning)
- For infants, **swaddling** or gently holding them can reduce physical movement and provide a calming effect.
- **Distraction during needle Insertion:** Encourage children to focus on a specific object, toy, or their favourite character during needle insertion.
- Prepare distraction tools such as **toys, games, music, or videos**. Electronic devices (such as tablets) with videos or interactive games can be helpful to keep the child's focus away from the procedure.
- Consider **Bubble-blowing** as a fun and effective distraction for younger children.
- Using toys, engaging in **pretend play**, or letting the child play with medical equipment (like toy syringes or stethoscopes) before the procedure can help normalise the process.
- Offering rewards (e.g. stickers) or praise for bravery after the procedure can help create positive associations with the experience.

References:

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- 3) Griffith RJ, Jordan V, Herd D, Reed PW, Dalziel SR. Vapocoolants (Cold Spray) for Pain Treatment During Intravenous Cannulation. *The Cochrane Database of Systematic Reviews*. 2016;4:CD009484. doi:10.1002/14651858.CD009484.pub2.
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- 5) Ghasemi M, Hoseinaliabadi P, Yazdanpanah F, et al. Comparison of Music and Vapocoolant Spray in Reducing the Pain of Venous Cannulation in Children Age 6-12: A Randomized Clinical Trial. *BMC Pediatrics*. 2022;22(1):237. doi:10.1186/s12887-022-03271-9.
- 6) Cohen LL. Behavioral Approaches to Anxiety and Pain Management for Pediatric Venous Access. *Pediatrics*. 2008;122 Suppl 3:S134-9. doi:10.1542/peds.2008-1055f.
- 7) Leahy S, Kennedy RM, Hesselgrave J, et al. On the Front Lines: Lessons Learned in Implementing Multidisciplinary Peripheral Venous Access Pain-Management Programs in Pediatric Hospitals. *Pediatrics*. 2008;122 Suppl 3:S161-70. doi:10.1542/peds.2008-1055i.
- 8) <https://www.nationwidechildrens.org/family-resources-education/health-wellness-and-safety-resources/helping-hands/comfort-hold-techniques>

Appendix A:

Examples of comfort positioning during venepuncture or intravenous cannula insertion:



Back-to-chest positioning:

- Child sits on parent's lap facing away from parent
- Child can turn head towards or away from procedure
- Child's arm goes under or over parent's arm
- Assistant to help hold patient's arm still



Chest-to-chest positioning:

- Child sits on parent's lap facing them
- Child's legs wrapped around parent's waist
- Child can turn head towards or away from procedure
- Child's arm goes under or over parent's arm
- Assistant to help hold patient's arm still

Appendix B:

Workflow for reducing pain and anxiety related to needle procedures in the Children's Emergency Department:

Early identification of child requiring intravenous cannulation/venepuncture.

Examples of such patients include:

- 1) Child with limb deformity who will likely require M&R under intravenous sedation
- 2) Child older than 5 years with deep laceration who will likely require T&S under intravenous sedation (Those unlikely to tolerate procedure with LA/LET gel)
- 3) Child with wound that will likely require intravenous antibiotics whilst in CE (e.g. patient with high likelihood of requiring referral to subspeciality such as HRM/Plastics team)
- 4) Consider for children who appear mild/moderately dehydrated who may require intravenous rehydration if unable to tolerate oral fluid challenge.

Preparation:

- 1) Identify appropriate veins for procedure and apply thick layer (approx. 1-2mm) of EMLA to area for cannulation/venepuncture. Best to apply to more than 1 site.
- 2) Cover area with Tegaderm.
- 3) Note time of application and allow contact with skin for at least 60 minutes.

Ask parents to monitor for any local/systemic reaction – redness/swelling/blistering.

Mild irritation/redness may occur which should settle once cream removed.

Performing procedure:

- 1) Remove cream and wipe clean.
- 2) Ensure child is in comfortable position – try to implement comfort positioning suggested in Appendix A where possible
- 3) Use distraction techniques in form of: Video/music/reading a book/bubbles/blowing pinwheel
- 4) Once procedure is complete, offer stickers as a reward